

Defect Report

Project: Quanta (Execution-as-a-Service Platform)

Module: AI-driven SightedPlanner

Field	Details
Defect ID	DEF-QTA-042
Title	SightedPlanner execution loop fails to parse dynamic DOM mutations in React-based Single Page Applications (SPAs)
Date Reported	May 27, 2026
Reporter	Muhammad Aliyan Nadeem
Severity	Critical
Priority	High
Environment	Staging (Node.js v20.x, Puppeteer v22.x, Ubuntu 22.04)
Description	When executing automated workflows on SPAs that utilize heavy dynamic DOM rendering, the SightedPlanner loses target node references after asynchronous network payloads resolve, resulting in an infinite execution loop and subsequent timeout.
Steps to Reproduce	1. Initialize an execution payload targeting https://[dummy-react-app].com/dashboard. 2. Instruct SightedPlanner to trigger a data-fetch modal. 3. Wait for the asynchronous DOM

	mutation. 4. Attempt to execute a click action on the newly rendered #submit-btn within the modal.
Expected Result	The SightedPlanner should detect the DOM mutation, re-evaluate the DOM tree, locate #submit-btn, and execute the click action within 500ms.
Actual Result	The SightedPlanner maintains the stale DOM snapshot, fails to locate #submit-btn, enters an infinite retry loop, and eventually throws a TargetNodeNotFound exception after the 30-second timeout threshold.
Root Cause Analysis (Preliminary)	The mutation observer integration within the SightedPlanner's execution loop is not successfully invalidating the cached DOM tree when specific asynchronous UI updates occur.
Status	Open

Updated Requirements Traceability Matrix (RTM)

The traceability matrix ensures that every requirement is tested and tracks any defects preventing the fulfillment of those requirements.

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Status	Defect ID	Defect Status
REQ-QTA-015	The system must execute automated clicks	TC-QTA-088	Verify interaction with elements loaded via	Passed	N/A	N/A

	on dynamic elements rendered asynchronously.		AJAX/Fetch on standard web pages.			
REQ-QTA-016	The SightedPlanner must maintain element targeting accuracy across state changes in SPAs.	TC-QTA-089	Verify workflow execution on React/Vue SPAs post-DOM mutation.	Failed	DEF-QTA-042	Open
REQ-QTA-017	The execution engine must timeout and log errors gracefully after 30 seconds of inactivity.	TC-QTA-090	Verify timeout handling for non-existent target nodes.	Passed	N/A	N/A

Review the defect and the mapping. Fix the mutation observer logic in the SightedPlanner before the execution engine requires a complete refactor.